MSDI Calculator
Crop Water Demand and Irrigation Requirement Methodology

Technical Methodology Memorandum

Prepared for internal model documentation and technical review
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Document element	Description
Model component	Seasonal crop water demand, effective rainfall, net irrigation requirement, mSDI-e runtime, and seasonal applied water volume.
Primary purpose	Document the equations and processing workflow used to convert seasonal weather and crop inputs into irrigation water requirements.
Primary data sources	Weather_LU, Crops_LU, Soils_LU, Coefficients_LU, and background_calculations TypeScript implementation files.
Output variables	Seasonal ETc, effective rainfall, irrigation requirement, weekly demand, runtime, total applied volume, supplemental water volume, and manure liquid volume.
Relationship to future scenarios	Future scenarios substitute scenario ET0 and precipitation into the same calculation framework; downstream equations are unchanged.

This memorandum separates two related but distinct concepts. Part I documents how weather and crop coefficients are used to estimate crop water demand. Part II documents how crop water demand is converted into irrigation requirement, runtime, and applied water volumes. This distinction is important because the crop water demand calculation estimates crop evapotranspiration, while the irrigation requirement calculation determines the quantity of water that must be supplied after accounting for effective rainfall and applied manure liquid.

Purpose and Scope

This memorandum documents the crop water demand and irrigation requirement methodology used in the MSDI Calculator. The methodology describes how seasonal reference evapotranspiration, seasonal precipitation, crop coefficients, soil rainfall effectiveness factors, mSDI-e system capacity assumptions, and manure liquid application volumes are combined to estimate irrigation demand and applied water volumes. The document is intended for model documentation, technical review, and eventual consolidation into a larger MSDI Calculator methodology manual.

The methodology applies to the screening-level mSDI-e analysis in the current application. It does not by itself document the source development of weather data, crop coefficients, soil coefficients, cost coefficients, manure coefficients, or environmental impact coefficients. Those subjects are documented or planned as separate methodology chapters. This memorandum focuses on the formulas and calculation sequence used by the application once those lookup values are available.

Terminology and Units

Term	Definition in this methodology	Primary units
ET0	Seasonal alfalfa-reference evapotranspiration used as the climate demand input.	inches per season
Kc	Crop coefficient used to convert reference evapotranspiration to crop water demand.	dimensionless
ETc	Crop water demand or crop evapotranspiration calculated as ET0 times Kc, with seasonal crop-transition weighting.	inches per season
Effective rainfall	Portion of seasonal precipitation treated as available to satisfy crop water demand.	inches per season
NIR	Net irrigation requirement. In the current code this value is named gross irrigation demand, but mathematically it is ETc minus effective rainfall before runtime compensation.	inches per season

Weekly demand	Seasonal irrigation requirement converted to an equivalent weekly depth.	inches per week
Runtime	Hours per week per set required to meet weekly demand under the mSDI-e system assumptions.	hours per week per set
Applied volume	Total seasonal volume applied by the mSDI-e system, including water and manure liquid.	gallons per season
Supplemental water volume	Water-only portion of total seasonal applied volume after accounting for manure liquid applied through the system.	gallons per season
Manure liquid volume	Liquid manure portion of total seasonal applied volume.	gallons per season

Source Lookup Tables and Implementation Inputs

The model uses lookup values prepared outside the crop water demand calculation. This section describes how those lookup tables enter the calculation, not how the lookup coefficients themselves were derived. Weather preparation is documented in the MSDI Calculator Weather Data Methodology and Metadata memorandum. Crop and soil coefficient development should be documented separately if source-specific coefficient justification is needed.

Lookup table	Relevant fields	Role in this methodology
Weather_LU	spring_eto, summer_eto, fall_eto, winter_eto; spring_precip, summer_precip, fall_precip, winter_precip; future RCP ET0 and precipitation fields	Provides seasonal weather values by county lookup key. Baseline values support current-condition calculations; RCP fields support future scenario analysis.
Crops_LU	Fall Growing Days, Winter Growing Days, Spring Growing Days, Summer Growing Days; Fall Kc, Winter Kc, Spring Kc, Summer Kc	Provides crop duration and seasonal crop coefficients used to calculate seasonal ETc.
Soils_LU	P_eff_Fall, P_eff_Winter, P_eff_Spring, P_eff_Summer; other soil-water and salt fields	Provides season-specific soil precipitation effectiveness factors and other soil-related fields used in downstream water and manure-application constraints.
Coefficients_LU / coefficients module	Days per season, days per week, inches per foot, gallons per acre-foot, DU compensation factor, system GPM, minutes per hour	Provides constants used to convert seasonal water demand into weekly demand, runtime, and gallons.
FormResults user inputs	County, crop selections, soil texture, acres, manure system selections, and system configuration values	Defines county weather key, crop rotation, soil category, acreage, and manure application conditions.

Conceptual Calculation Framework

The calculation sequence intentionally separates crop demand from irrigation supply. Weather and crop coefficients are used first to calculate crop water demand. Effective rainfall and applied liquid manure are then treated as sources of water that can satisfy a portion of that demand. Supplemental irrigation water is the remaining water required to meet the seasonal crop water requirement after accounting for those sources.

Part	Primary calculation sequence	Primary output
Part I - Crop water demand	Seasonal ET0 and crop coefficients -> seasonally weighted ETc	Seasonal crop water demand (ETc)
Part II - Irrigation requirement and applied volume	ETc, effective rainfall, manure liquid, system runtime, and system flow rate -> applied volumes	Seasonal irrigation requirement, runtime, total applied volume, water-only volume, and manure-liquid volume

Part I - Crop Water Demand

1.1 Seasonal Reference Evapotranspiration Inputs

The model retrieves seasonal ET0 from Weather_LU using the county lookup key constructed from the selected county. Baseline seasonal ET0 fields are spring_eto, summer_eto, fall_eto, and winter_eto. Future scenario fields use the same season structure with RCP and time-point suffixes. The weather methodology documents the preparation of baseline ET0 and future ET0 scenario values.

The use of ET0 and Kc follows the standard crop coefficient framework described in FAO Irrigation and Drainage Paper 56, where climate demand is represented by reference evapotranspiration and crop-specific differences are represented by crop coefficients [1]. The MSDI workflow uses alfalfa-reference ET0, consistent with the weather dataset documentation and the project decision to use an agricultural reference ET basis.

1.2 Seasonal Crop Water Demand (ETc)

The model calculates seasonal crop water demand using a crop coefficient approach. The base relationship is:

Equation 1. $ETc = ET0 \times Kc$

where ETc is crop water demand, ET0 is seasonal alfalfa-reference evapotranspiration, and Kc is the crop coefficient for the crop and season. The implementation extends this base relationship by allowing a season to include a transition from one crop to the next. Seasonal crop water demand is therefore calculated as a weighted average of two crop-specific ETc values within the same season.

Equation 2. $\text{Seasonal ETc} = (D\_primary / D\_season) \times (Kc\_primary,season \times ET0\_season) + ((D\_season - D\_primary) / D\_season) \times (Kc\_next,season \times ET0\_season)$

where D_primary is the number of growing days assigned to the primary crop within the season, D_season is the model season length, Kc_primary,season is the primary crop coefficient for the season, and Kc_next,season is the seasonal Kc for the following crop in the rotation. This structure avoids assuming that a single crop occupies an entire season when the user-entered rotation implies seasonal transition.

Season	Primary crop in calculation	Next crop used for remaining days	ET0 field
Spring	Spring crop	Summer crop	spring_eto
Summer	Summer crop	Fall crop	summer_eto

Fall	Fall crop	Winter crop	fall_eto
Winter	Winter crop	Spring crop	winter_eto

1.3 Implementation Equations by Season

The equations below reproduce the implemented TypeScript logic in algebraic form. These equations are the primary technical basis for the crop water demand portion of the methodology.

Equation 3. Spring ETc = (SpringDays_springCrop / D) x (SpringKc_springCrop x SpringET0) + ((D - SpringDays_springCrop) / D) x (SpringKc_summerCrop x SpringET0)

Equation 4. Summer ETc = (SummerDays_summerCrop / D) x (SummerKc_summerCrop x SummerET0) + ((D - SummerDays_summerCrop) / D) x (SummerKc_fallCrop x SummerET0)

Equation 5. Fall ETc = (FallDays_fallCrop / D) x (FallKc_fallCrop x FallET0) + ((D - FallDays_fallCrop) / D) x (FallKc_winterCrop x FallET0)

Equation 6. Winter ETc = (WinterDays_winterCrop / D) x (WinterKc_winterCrop x WinterET0) + ((D - WinterDays_winterCrop) / D) x (WinterKc_springCrop x WinterET0)

In these equations, D is the model constant DAYS_IN_A_SEASON. The spring calculation uses the selected spring crop growing days and spring Kc, then applies the summer crop spring Kc to the remaining portion of the season. The same transition logic is applied cyclically for summer, fall, and winter.

Part II - Irrigation Requirement and Applied Water

2.1 Effective Rainfall

Effective rainfall is the portion of seasonal precipitation that is treated as available to meet crop water demand. The model calculates effective rainfall by multiplying seasonal precipitation by a soil-specific precipitation effectiveness factor.

Equation 7. Pe_season = P_season x f_Peff,soil

where Pe_season is effective rainfall, P_season is seasonal precipitation from Weather_LU, and f_Peff,soil is the soil-specific precipitation effectiveness factor from Soils_LU. The current implementation retrieves the season-specific effective rainfall field from Soils_LU for each season: P_EFF_SPRING for spring, P_EFF_SUMMER for summer, P_EFF_FALL for fall, and P_EFF_WINTER for winter. This reflects the corrected seasonal effective rainfall mapping and allows soil rainfall effectiveness to vary by season.

Season	Precipitation field	Effective rainfall coefficient used	Implemented equation
Spring	spring_precip	Soils_LU.P_EFF_SPRING	Spring Pe = spring_precip x P_EFF_SPRING
Summer	summer_precip	Soils_LU.P_EFF_SUMMER	Summer Pe = summer_precip x P_EFF_SUMMER
Fall	fall_precip	Soils_LU.P_EFF_FALL	Fall Pe = fall_precip x P_EFF_FALL
Winter	winter_precip	Soils_LU.P_EFF_WINTER	Winter Pe = winter_precip x P_EFF_WINTER

2.2 Irrigation Requirement

After ETc and effective rainfall are calculated, the model calculates the seasonal irrigation requirement as crop water demand minus effective rainfall. In the TypeScript implementation and legacy spreadsheet row labels, this value is named gross irrigation demand. Mathematically, however, it is equivalent to a seasonal net irrigation requirement because it is calculated before distribution-uniformity compensation and before conversion to runtime.

Equation 8. NIR_season = ETc_season - Pe_season

Implemented row	Software label	Equation
Row 36	Expected Spring Gross Irrigation Demand	Spring NIR = Spring ETc - Spring Pe
Row 37	Expected Summer Gross Irrigation Demand	Summer NIR = Summer ETc - Summer Pe
Row 38	Expected Fall Gross Irrigation Demand	Fall NIR = Fall ETc - Fall Pe
Row 39	Expected Winter Gross Irrigation Demand	Winter NIR = Winter ETc - Winter Pe

Because the implemented row names use gross irrigation demand, this memorandum uses the term "implementation GIR" when referring to the named code variables and uses "NIR" when referring to the hydrologic interpretation of the equation. The actual gross-up for distribution uniformity occurs later in the runtime calculation, not in Rows 36 through 39.

2.3 Contribution of Manure Liquid to Crop Water Supply

A key nuance of the MSDI Calculator is that the applied liquid stream can include both supplemental irrigation water and manure liquid. The water or moisture content of manure liquid contributes to crop water supply when it is applied through the mSDI-e system. Therefore, the model distinguishes between total liquid volume applied, manure liquid volume applied, and supplemental water-only volume applied.

The runtime and total volume calculations estimate the total seasonal liquid volume required to satisfy the irrigation requirement. A portion of that volume may be delivered as manure liquid, depending on the manure blend percentage determined by nutrient, salt, phosphorus, nitrogen, total dissolved solids, and filter-related limits in other background calculations. The remainder is supplemental irrigation water.

Equation 9. ManureVolume_weekly,season = ManureBlendPct_season x TotalAppliedVolume_weekly,season

Equation 10. WaterOnlyVolume_season = TotalAppliedVolume_season - ManureVolume_weekly,season x (D_season / 7)

Equation 11. ManureLiquidVolume_season = TotalAppliedVolume_season - WaterOnlyVolume_season

2.4 Manure Application Constraint Linkage

The manure-liquid contribution described above is not an unconstrained volume. In the MSDI Calculator, the manure blend percentage is determined in the manure nutrient and application mass-balance workflow before the seasonal water-only volume is finalized. This linkage is important because the amount of liquid manure that can be applied through mSDI-e is constrained by both crop nutrient demand and manure quality limitations.

At a conceptual level, the manure mass-balance module calculates the manure liquid stream composition at the terminus in the manure management system (ie: after collection, storage, digestion, separation, filtration, and precipitation contributions). The resulting seasonal manure composition includes total solids, TAN, TKN, TN, phosphorus, potassium, and total dissolved solids. The model then evaluates the maximum allowable manure application against limiting criteria, including salt/TDS constraints and TN constraints.

Equation 9A. MaxManure_salt,season = function(SeasonalSaltLimit, TDS_season, TotalAppliedVolume_season)

Equation 9B. BlendPct_salt,season = MaxManure_salt,season / TotalAppliedVolume_season

Equation 9C. MaxManure_N,season = function(SeasonalNDemand, TN_season, plant-available fraction, TotalAppliedVolume_season)

Equation 9D. BlendPct_N,season = MaxManure_N,season / TotalAppliedVolume_season

Equation 9E. ManureBlendPct_season = min(BlendPct_salt,true,season, BlendPct_N,true,season)

The equations above summarize the linkage between the irrigation requirement methodology and the manure application mass-balance methodology. The detailed salt-limited, N-limited, and true blend percentage calculations are documented in the MSDI Calculator Manure Nutrient and Application Mass Balance Methodology. In this irrigation methodology, the important point is that manure liquid volume is treated as water applied to the crop only after the allowable manure fraction has been constrained by the manure mass-balance module.

Therefore, total seasonal liquid application is partitioned into manure liquid and supplemental water as follows:

Equation 9F. ManureLiquidVolume_season = TotalAppliedVolume_season x ManureBlendPct_season

Equation 9G. SupplementalWaterOnlyVolume_season = TotalAppliedVolume_season - ManureLiquidVolume_season

This correction prevents the irrigation methodology from implying that manure-liquid application is chosen solely by water demand. The water value of manure is credited only to the extent that the manure stream can be applied without exceeding salt/TDS and TN constraints.

This treatment is important because a system applying large volumes of liquid manure may satisfy a meaningful portion of seasonal crop water demand through manure-derived water. In the results and economic calculations, the water-only volume is the supplemental irrigation water volume after crediting manure liquid water toward the total seasonal applied volume.

2.5 Weekly Irrigation Requirement

Seasonal irrigation requirement is converted to an equivalent weekly requirement so that system runtime can be calculated on a weekly operating basis. The conversion is identical for all four seasons.

Equation 12. WeeklyDemand_season = (NIR_season / D_season) x 7

Rows 88 through 91 implement this conversion for spring, summer, fall, and winter. The weekly value is expressed in inches per week and becomes the demand input for the runtime calculation.

2.6 Runtime Calculation

The runtime calculation converts weekly irrigation depth into the number of hours per week per set required to apply that volume with the assumed mSDI-e system. The calculation uses acreage per set, the system flow rate, a distribution-uniformity compensation factor, and standard water-volume conversions.

Equation 13. GallonsPerAcre_season = (WeeklyDemand_season / 12) x GallonsPerAcreFoot

Equation 14. GallonsPerSet_season = GallonsPerAcre_season x AcresPerSet

Equation 15. AdjustedGallonsPerSet_season = GallonsPerSet_season x DUCompensationFactor

Equation 16. RuntimeHoursPerWeekPerSet_season = AdjustedGallonsPerSet_season / SystemGPM / 60

Variable	Implementation source or constant	Description
WeeklyDemand_season	Rows 88-91	Equivalent weekly demand in inches per week.
GallonsPerAcreFoot	Coefficients module / Coefficients_LU	Volume conversion for one acre-foot.
AcresPerSet	Expected max acres per set calculation	Acreage served by one irrigation set.
DUCompensationFactor	Coefficients module / Coefficients_LU	Multiplier used to compensate for distribution uniformity.
SystemGPM	Coefficients module / Coefficients_LU	mSDI-e system flow rate in gallons per minute.
60	Minutes per hour	Converts minutes to hours.

2.7 Weekly and Seasonal Applied Volume

Weekly total applied volume is calculated from the number of sets, runtime, system flow rate, and minutes per hour.

Equation 17. TotalAppliedVolume_weekly,season = NumberOfSets x RuntimeHoursPerWeekPerSet_season x SystemGPM x 60

Seasonal total applied volume is then calculated by scaling weekly volume to the model season length.

Equation 18. TotalAppliedVolume_season = max(TotalAppliedVolume_weekly,season, 0) / 7 x D_season

Rows 146 through 149 implement this conversion for spring, summer, fall, and winter. The max operation reflects the spreadsheet-style IF condition in the implementation: negative weekly volumes are not carried forward into seasonal application volumes. This prevents negative seasonal applied-water outputs when effective rainfall exceeds calculated crop water demand.

Downstream Use of Calculated Water Values

The crop water demand and irrigation requirement calculations feed several downstream model components. The same seasonal water quantities are used to support mSDI-e runtime estimates, water-use reporting, water-cost calculations, scenario analysis, and future environmental modules (planned in future versions of the app.

Calculated output	Downstream use
Seasonal ETc	Foundation for irrigation requirement, climate scenario sensitivity, and future water-balance tables.
Effective rainfall	Reduces irrigation requirement and supports climate scenario water-balance outputs.
NIR / implementation GIR	Primary water requirement before runtime and DU compensation.
Weekly demand	Basis for runtime calculation.
Runtime hours/week/set	Basis for system operation estimates and total applied volume.
Total seasonal applied volume	Basis for total mSDI-e liquid volume delivered.
Water-only seasonal volume	Basis for irrigation water assessment and water savings calculations.
Manure liquid seasonal volume	Basis for manure nutrient delivery and future environmental calculations.

Manure blend percentage	Determines how much of total seasonal applied liquid is supplied as manure liquid versus supplemental irrigation water; derived from salt/TDS and TN constraints in the manure mass-balance module.

Relationship to Future Climate Scenario Modeling

Future climate scenario analysis will use the same methodology described in this memorandum. The scenario layer should substitute future ET0 and future precipitation values for the baseline seasonal weather values and then run the same crop water demand, effective rainfall, irrigation requirement, runtime, and applied-volume calculations. This preserves comparability between baseline and RCP scenario outputs.

Equation 19. ETc_RCP,time,season = ET0_RCP,time,season x Kc_season, with the same crop-transition weighting used for baseline

Equation 20. NIR_RCP,time,season = ETc_RCP,time,season - Pe_RCP,time,season

No RCP-specific water-demand equation is required. The scenario analysis is an input-substitution workflow: scenario ET0 and precipitation replace baseline ET0 and precipitation, while crop coefficients, soil factors, runtime conversions, and applied-volume equations remain unchanged unless a future model version explicitly modifies them.

Assumptions and Limitations

- The model uses seasonal rather than daily water-balance calculations. It does not simulate daily soil-water depletion, rainfall timing, runoff, or percolation events.
- Crop water demand is estimated using lookup-table Kc values and seasonal ET0, not crop-stage-specific daily Kc curves.
- Seasonal crop transitions are represented by growing-day weighting between two crops within each season.
- Effective rainfall is calculated using soil- and season-specific coefficients. The implementation should use P_EFF_SPRING, P_EFF_SUMMER, P_EFF_FALL, and P_EFF_WINTER for the corresponding seasons.
- The variables named gross irrigation demand in the implementation are mathematically equivalent to net irrigation requirement before distribution-uniformity compensation.
- Distribution-uniformity compensation is applied during runtime conversion rather than directly in Rows 36 through 39.
- Manure liquid is treated as a liquid water source when applied through the system. The model separates total liquid volume into manure liquid volume and supplemental water-only volume.
- Runtime and applied-volume estimates are screening-level calculations and should not be interpreted as irrigation scheduling recommendations.
- Field-level design should use site-specific soil, crop, hydraulic, water quality, and irrigation-system information where available.

Quality Control and Validation Checks

Check	Purpose
Confirm Weather_LU ET0 and precipitation units	Avoid mixing inches, millimeters, seasonal totals, and seasonal means.
Confirm crop Kc fields align with season names	Prevent use of winter Kc in summer calculations or other season mismatches.
Confirm growing-day values are within 0 to DaysPerSeason	Avoid negative crop-transition weights or weights greater than 1.
Confirm effective rainfall factor used intentionally	Confirm each season uses the corresponding seasonal coefficient: P_EFF_SPRING, P_EFF_SUMMER, P_EFF_FALL, and P_EFF_WINTER.
Check for negative NIR cases	Where effective rainfall exceeds ETc, downstream seasonal volume should be zero after max/IF handling.
Check manure volume partitioning	Total seasonal volume should equal water-only volume plus manure-liquid volume.
Compare sample counties manually	Verify ET0, precipitation, ETc, effective rainfall, and seasonal applied volume for representative dairy regions.
Validate future scenario substitution	Scenario runs should change only weather inputs unless a separate model change is intentionally implemented.

Processing Workflow Summary

1. Build county lookup key from the selected county and state.
2. Retrieve seasonal ET0 and precipitation values from Weather_LU.
3. Retrieve crop growing days and seasonal Kc values from Crops_LU.
4. Calculate seasonally weighted ETc for spring, summer, fall, and winter.
5. Retrieve the season-specific soil effective rainfall factor from Soils_LU.
6. Calculate seasonal effective rainfall.
7. Calculate seasonal NIR as ETc minus effective rainfall.
8. Convert seasonal NIR to weekly demand in inches per week.
9. Convert weekly demand to gallons per acre and gallons per set.
10. Apply the DU compensation factor.
11. Convert adjusted gallons to runtime hours per week per set.
12. Calculate weekly total applied volume from runtime, number of sets, system flow rate, and minutes per hour.
13. Scale weekly volume to seasonal volume.
14. Determine the allowable manure blend using the manure mass-balance constraints, then partition seasonal volume into supplemental water-only volume and manure-liquid volume.
15. Pass calculated water volumes to reporting, economic, and future environmental modules.

Implementation Traceability

Calculation stage	Implementation files or rows
Crop water demand	Rows 28-31: expected seasonal crop water demand.
Effective rainfall	Rows 32-35: expected seasonal effective rainfall.
Irrigation requirement	Rows 36-39: expected seasonal gross irrigation demand, mathematically equivalent to NIR.
Weekly demand	Rows 88-91: seasonal demand converted to weekly inches.
Runtime	Rows 94-97: weekly demand converted to hours per week per set.
Weekly applied volume	Rows 98-101: runtime converted to gallons per week.
Seasonal applied volume	Rows 146-149: weekly volume converted to seasonal gallons.
Water-only volume	Rows 150-153: total seasonal volume minus seasonalized manure liquid gallons.
Manure liquid volume	Rows 154-157: total seasonal volume minus water-only volume.

References and Source Links

[1] Allen, R. G., Pereira, L. S., Raes, D., and Smith, M. 1998. Crop evapotranspiration - Guidelines for computing crop water requirements. FAO Irrigation and Drainage Paper 56. https://www.fao.org/4/X0490E/x0490e00.htm

[2] ASCE Environmental and Water Resources Institute. The ASCE Standardized Reference Evapotranspiration Equation. ASCE-EWRI Task Committee documentation. https://www.mesonet.org/images/site/ASCE_Evapotranspiration_Formula.pdf

[3] MSDI Calculator Weather Data Methodology and Metadata. Internal technical memorandum, 2026.

[4] MSDI Calculator lookup tables. Weather_LU, Crops_LU, Soils_LU, and Coefficients_LU workbook, 2026.

[5] MSDI Calculator TypeScript implementation. src/lib/background_calculations rows 28-39, 88-91, 94-101, and 146-157.

[6] MSDI Calculator Manure Nutrient and Application Mass Balance Methodology. Internal technical memorandum, 2026.

Appendix A. Key Implementation Equations

This appendix consolidates the primary equations used in the current calculation workflow.

A-1. $\mathrm{ETc} = \mathrm{ET0} \times \mathrm{Kc}$

A-2. $\text{Seasonal ETc} = (D_{primary} / D_{season}) \times (\mathrm{Kc}_{primary} \times \mathrm{ET0}) + ((D_{season} - D_{primary}) / D_{season}) \times (\mathrm{Kc}_{next} \times \mathrm{ET0})$

A-3. $\mathrm{Pe} = P \times f_{Peff,soil}$

A-4. $\mathrm{NIR} = \mathrm{ETc} - \mathrm{Pe}$

A-5. $\mathrm{WeeklyDemand} = \mathrm{NIR} / D_{season} \times 7$

A-6. $\mathrm{GallonsPerAcre} = \mathrm{WeeklyDemand} / 12 \times \mathrm{GallonsPerAcreFoot}$

A-7. $\mathrm{GallonsPerSet} = \mathrm{GallonsPerAcre} \times \mathrm{AcresPerSet}$

A-8. $\mathrm{AdjustedGallonsPerSet} = \mathrm{GallonsPerSet} \times \mathrm{DUCompensationFactor}$

A-9. $\mathrm{RuntimeHoursPerWeekPerSet} = \mathrm{AdjustedGallonsPerSet} / \mathrm{SystemGPM} / 60$

A-10. $\mathrm{TotalAppliedVolume}_{weekly} = \mathrm{NumberOfSets} \times \mathrm{RuntimeHoursPerWeekPerSet} \times \mathrm{SystemGPM} \times 60$

A-11. $\mathrm{TotalAppliedVolume}_{season} = \max(\mathrm{TotalAppliedVolume}_{weekly}, 0) / 7 \times D_{season}$

A-12. $\mathrm{WaterOnlyVolume}_{season} = \mathrm{TotalAppliedVolume}_{season} - \mathrm{ManureVolume}_{weekly} \times D_{season} / 7$

A-13. $\mathrm{ManureLiquidVolume}_{season} = \mathrm{TotalAppliedVolume}_{season} - \mathrm{WaterOnlyVolume}_{season}$

A-14. $\mathrm{BlendPct}_{salt} = \mathrm{MaxManure}_{salt} / \mathrm{TotalAppliedVolume}$

A-15. $\mathrm{BlendPct}_{N} = \mathrm{MaxManure}_{N} / \mathrm{TotalAppliedVolume}$

A-16. $\mathrm{ManureBlendPct} = \min(\mathrm{BlendPct}_{salt\_true}, \mathrm{BlendPct}_{N\_true})$

A-17. $\mathrm{SupplementalWaterOnlyVolume} = \mathrm{TotalAppliedVolume} - \mathrm{ManureLiquidVolume}$